Francesc Comalada i Pla

Predoctoral Student
Catalan Institute for Water Research (ICRA)

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Research Interests

Spatio-temporal modelling; global change ecology; freshwater ecosystems; geoinformatics; biodiversity conservation.

Education

Ph.D., Water Science and Technology 2024 – present
University of Girona & Catalan Institute for Water Research (ICRA)
Thesis: Bayesian Inference of Recreation–Conservation Trade-offs in Freshwater Ecosystems. Evidence from Social Media Data
Supervisors: Dr. Vicenç Acuña; Dr. Xavier Garcia

M.Sc., Environmental Change 2021 – 2022
University of Girona
Thesis: Using Twitter Geotagged Posts to Map the Cultural Ecosystem Services of Freshwater Landscapes

B.Sc., Geography, Land Planning and Environmental Management 2017 – 2021
University of Girona
Thesis: Assessing Ecosystem Services in the Tordera's Delta Following a Flooding Event: From Risks to Opportunities

Publications

Published

1. **Comalada**, **F.**, Acuña, V., & Garcia, X. (2025c). Modelling cultural ecosystem services of river landscapes in the Iberian Peninsula with deep learning and social media images. *Journal of Environmental Management*, *394*, 127667. DOI: 10.1016/j.jenvman.2025.127667.
2. **Comalada**, **F.**, Acuña, V., & Garcia, X. (2026). FreshCES-Net: A scalable deep learning approach to map freshwater cultural ecosystem services using social media data. *SoftwareX*, *33*, 102502. DOI: 10.1016/j.softx.2025.102502.
3. **Comalada**, **F.**, Llorente, O., Acuña, V., Saló, J., & Garcia, X. (2025a). Using georeferenced text from social media to map the cultural ecosystem services of freshwater ecosystems. *Ecosystem Services*, *72*, 101702. DOI: 10.1016/j.ecoser.2025.101702.

Under Review

1. **Comalada**, **F.**, Acuña, V., & Garcia, X. (2025a). Cultural ecosystem services of inland aquatic ecosystems: A systematic review of methods, biophysical drivers, and management implications [Under review]. *People and Nature*.
2. **Comalada**, **F.**, Acuña, V., Serra, L., & Garcia, X. (2026). Spatiotemporal modelling of post-COVID freshwater recreation reveals persistent and emerging pressures [Under review]. *Journal of Outdoor Recreation and Tourism*.
3. **Comalada**, **F.**, Garcia, X., Sinclair, M., & Acuña, V. (2026a). Monetary valuation of cultural ecosystem services to support river restoration in Europe [Under review]. *Ecological Economics*.
4. **Comalada**, **F.**, Garcia, X., Toivonen, T., & Acuña, V. (2026). Do recreationists care about ecological quality? Evidence from Mediterranean freshwater ecosystems [Under review]. *One Earth*.

Conference Contributions

1. **Comalada**, **F.**, Acuña, V., & Garcia, X. (2025b). Mapping cultural ecosystem services of river landscapes in the Iberian Peninsula using artificial intelligence and social media (oral communication). *IALE 2025 European Landscape Ecology Congress, Bratislava, Slovakia*.
2. **Comalada**, **F.**, Acuña, V., & Garcia, X. (2025d). Modelling ecosystem services of rivers in the iberian peninsula using social media data (oral communication). *III Edition Divulga NextGen*.
3. **Comalada**, **F.**, Acuña, V., & Garcia, X. (2025e). Policy implications of cultural ecosystem services in freshwater ecosystems: A systematic review (poster). *IALE 2025 European Landscape Ecology Congress, Bratislava, Slovakia*.
4. **Comalada**, **F.**, Garcia, X., Sinclair, M., & Acuña, V. (2026b). Monetary valuation of cultural ecosystem services to support river restoration in Europe (oral communication). *VI Annual International Congress of Doctoral Students (CAIED), Miguel Hernández University. Online*.
5. **Comalada**, **F.**, Llorente, O., Acuña, V., Saló, J., & Garcia, X. (2025b). Using georeferenced text from social media to map the cultural ecosystem services of freshwater ecosystems (oral communication). *5th Annual International Congress of Doctoral Students, Miguel Hernández University*.
6. **Comalada**, **F.**, Saló, J., & Garcia, X. (2022). Using twitter geotagged posts to map the cultural ecosystem services of freshwater landscapes (poster). *4th ESP Europe Conference 2022, Heraklion, Greece*.

Research Experience

PhD Candidate, Catalan Water Research Institute (ICRA) 2024 – present

Thesis: Bayesian Inference of Recreation–Conservation Trade-offs in Freshwater Ecosystems.
Supervisors: Dr. Vicenç Acuña; Dr. Xavier Garcia

- Developed Bayesian models to quantify recreation–conservation trade-offs using large-scale social media geodata.
- Designed and implemented deep learning pipelines for cultural ecosystem service mapping across river landscapes.
- Framed within the EU MERLIN project (Grant No. 101036337), focused on freshwater ecosystem restoration across Europe.

Visiting Researcher, University of Helsinki Aug – Dec 2025

Digital Geography Lab. Supervisor: Dr. Tuuli Toivonen

- Collaborated on spatiotemporal modelling of outdoor recreation using volunteered geographic information and social media data.

Research Internships

Researcher, Catalan Water Research Institute (ICRA) May – Oct 2022
MERLIN Project (No. 101036337)

Researcher, Environmental Institute (IME) Jun – Oct 2022
Sub-directorate General of Maritime Policy, Govern of Catalonia

Researcher, University of Girona Feb – May 2022
Laboratory of Landscape Analysis and Management

Researcher, Environmental Institute (IME) Jun – Oct 2021
Sub-directorate General of Maritime Policy, Govern of Catalonia

Technician, Fornells de la Selva Townhall Jan – Jun 2021
Fornells de la Selva, Girona

Skills

Programming	Python, R, SQL, LaTeX
GIS	QGIS, ArcGIS, PostGIS
Version control	Git, GitHub
Languages	Catalan (native), Spanish (native), English (C1)

Certifications

2025	Bayesian Methods for Environmental Modelling – Summer School. EAWAG
2025	Exploring Nature-Based Tourism and Visitor Monitoring in Theory and Practice – Summer School. University of Helsinki
2024	Certificate in Advanced English C1. Cambridge University
2024	Using Python for Research. Harvard University
2024	GIS Programming for Environmental Research with Open-Source Software (QGIS, R, Python). Barcelona Institute for Global Health
2023	Statistical Learning with Python. StanfordOnline
2023	Introduction to PostgreSQL and PostGIS. Barcelona Activa
2022	Spanish Official Drone Pilot Training. AEROFOR

Outreach Activities

2025	Workshop RE4GREEN: Supporting the Transition to a Sustainable Economy and Society. UAB
2025	Professional Orientation Day, Faculty of Sciences, University of Girona
2025	Outreach on water consumption in collaboration with Basket Girona
2024	European Night of Research (NitRecerCat2425, Grant No. 101162003)
2023	Oral communication at Pint of Science: *AI for Mapping Cultural Services with Twitter*

Awards & Achievements

2024	Joan Oró Predoctoral Fellowship. Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR), Generalitat de Catalunya
2023	Excellency Grant. UdG, Santander Bank and Social Council of the University of Girona

References

Available upon request.